

17 Comparator (COMP)

17.1 Introduction

STM32F303xB/C/D/E, STM32F358xx and STM32F398xx embed seven general purpose comparators that can be used either as standalone devices (all terminal are available on I/Os) or combined with the timers. STM32F303x6/8 and STM32F328x8 embed three comparators, COMP2, COMP4 and COMP6.

The comparators can be used for a variety of functions including:

- Wake-up from low-power mode triggered by an analog signal,
- Analog signal conditioning,
- Cycle-by-cycle current control loop when combined with the DAC and a PWM output from a timer.

17.2 COMP main features

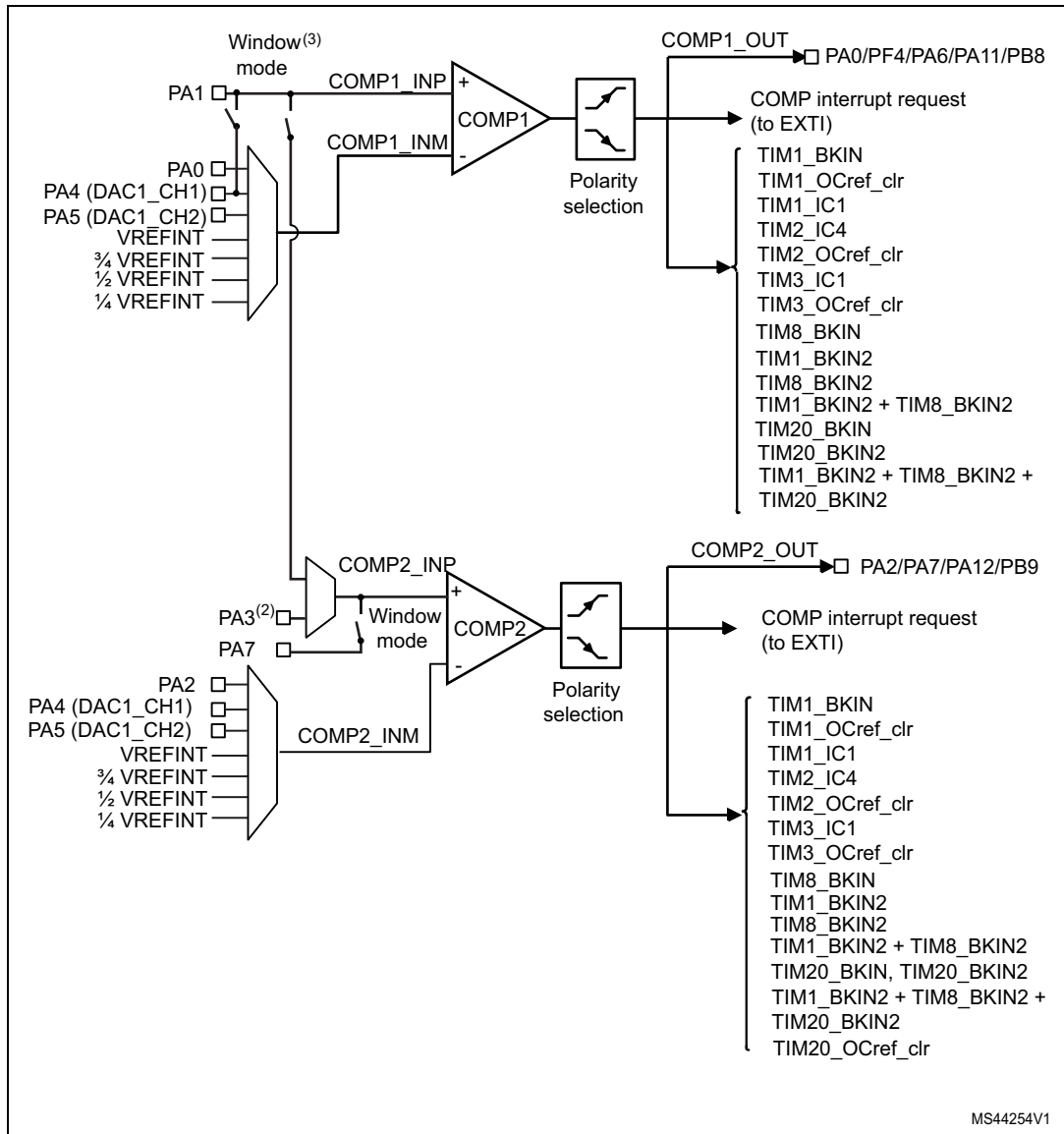
- Rail-to-rail comparators
- Each comparator has positive and configurable negative inputs used for flexible voltage selection:
 - Multiplexed I/O pins
 - DAC1 channel 1, DAC1 channel 2, DAC2 channel1 on STM32F303x6/8 and STM32F328x8 devices and DAC1 channel 1, DAC1 channel 2 on STM32F303xB/C/D/E, STM32F358xx and STM32F398xx devices.
 - Internal reference voltage and three submultiple values (1/4, 1/2, 3/4) provided by scaler (buffered voltage divider)
- Programmable hysteresis (on STM32F303xB/C and STM32F358xC only)
- Programmable speed / consumption (on STM32F303xB/C and STM32F358xC only)
- The outputs can be redirected to an I/O or to timer inputs for triggering:
 - Capture events
 - OCREF_CLR events (for cycle-by-cycle current control)
 - Break events for fast PWM shutdowns
- COMP1/COMP2, COMP3/COMP4 and COMP5/COMP6 comparators can be combined in a window comparator. This applies to STM32F303xB/C and STM32F358xC devices only. COMP7 does not support the window mode.
- Comparator outputs with blanking source
- Each comparator has interrupt generation capability with wake-up from Sleep and Stop modes (through the EXTI controller)

17.3 COMP functional description

17.3.1 COMP block diagram

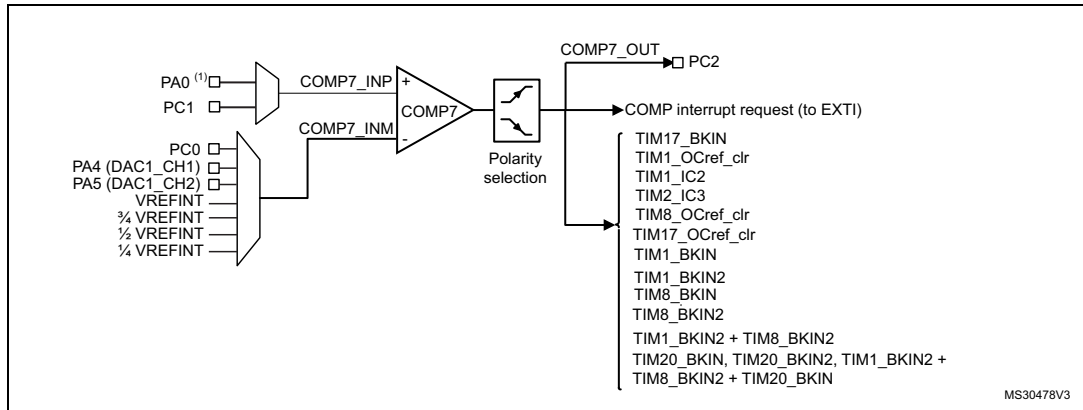
The block diagram of the comparators is shown in [Figure 122: Comparator 1 and 2 block diagrams \(STM32F303xB/C/D/E, STM32F358xC and STM32F398xE\)](#) and [Figure 123: STM32F303xB/C/D/E, STM32F358xC and STM32F398xE comparator 7 block diagram](#).

Figure 122. Comparator 1 and 2 block diagrams (STM32F303xB/C/D/E, STM32F358xC and STM32F398xE)



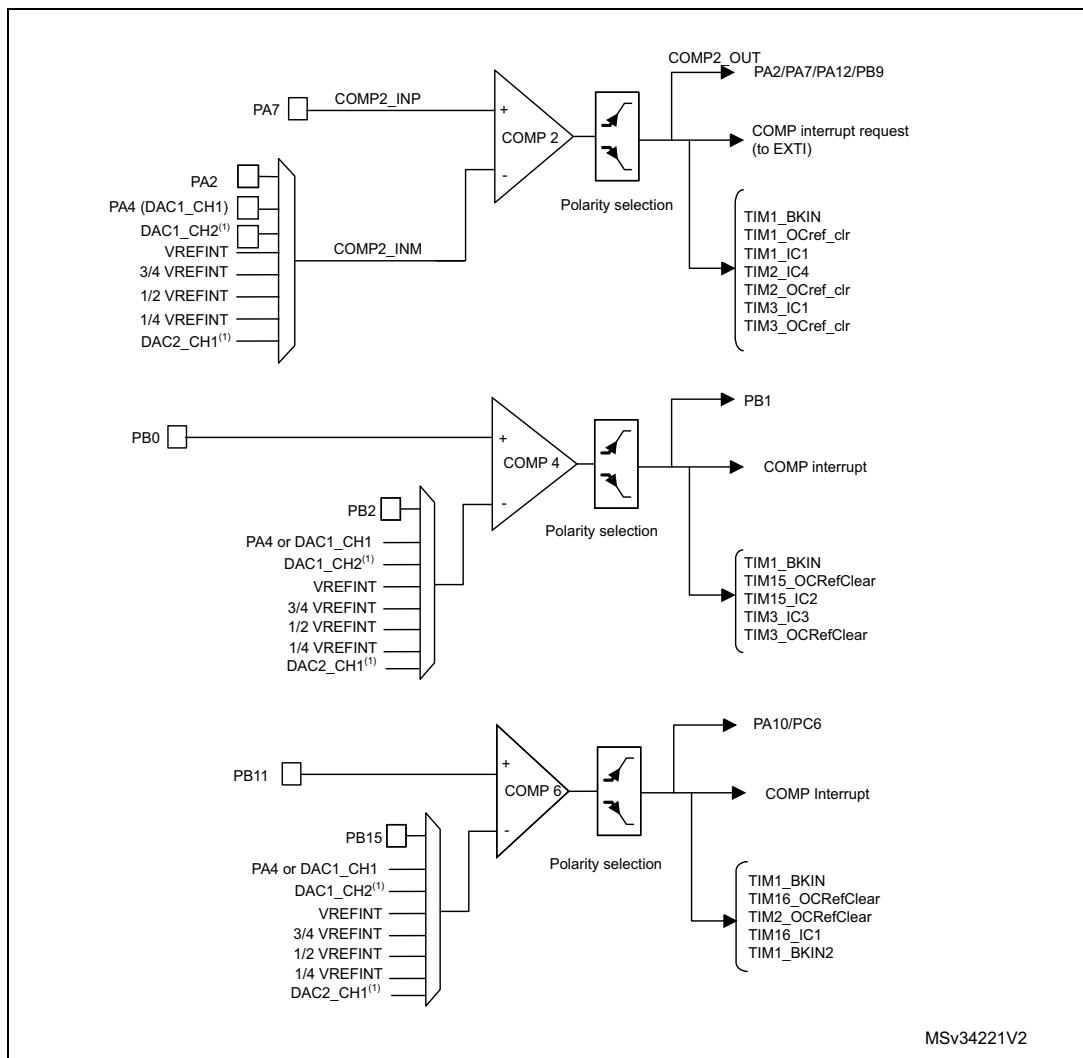
1. For a complete block diagram of comparators 1 to 6, refer to [Section 18: Operational amplifier \(OPAMP\)](#), where all block diagrams and interconnections between comparators 1 to 6 and operational amplifiers are given.
2. Only on STM32F303xB/C and STM32F358xC.
3. Window mode is not supported in STM32F303xD/E and STM32F398xE.

Figure 123. STM32F303xB/C/D/E, STM32F358xC and STM32F398xE comparator 7 block diagram



1. Only for STM32F303xB/C and STM32F358xC

Figure 124. STM32F303x6/8 and STM32F328x8 comparators 2/4/6 block diagrams



1. In STM32F303x6/8 and STM32F328x8 devices, DAC1_CH2 and DAC2_CH1 outputs are connected

directly, thus PA5 and PA6 are not available as COMPx_INM (x = 2,4,6) inputs. When DAC1_OUT2 and DAC2_OUT1 are connected internally to comparator non inverting input, the I/Os on which the DAC1_OUT and DAC2_OUT1 are mapped (PA5 and PA6) can be used as GPIOs.

17.3.2 COMP pins and internal signals

The I/Os used as comparators inputs must be configured in analog mode in the GPIOs registers.

The comparator output can be connected to the I/Os using the alternate function channel given in “Alternate function mapping” table in the datasheet.

The table below summarizes the I/Os that can be used as comparators inputs and outputs.

The output can also be internally redirected to a variety of timer input for the following purposes:

- Emergency shut-down of PWM signals, using BKIN and BKIN2 inputs
- Cycle-by-cycle current control, using OCREF_CLR inputs
- Input capture for timing measures

It is possible to have the comparator output simultaneously redirected internally and externally.

Table 108. Comparator input/output summary

	Comparator inputs/outputs						
	COMP1	COMP2	COMP3	COMP4	COMP5	COMP6	COMP7 ⁽³⁾
Comparator inverting input: connection to internal signals	DAC1_CH1 DAC1_CH2 DAC2_CH1 ⁽¹⁾ Vrefint $\frac{3}{4}$ Vrefint $\frac{1}{2}$ Vrefint $\frac{1}{4}$ Vrefint						
Comparator inputs connected to I/Os (+: non inverting input; -: inverting input)	+: PA1 -: PA0	+: PA3 ⁽²⁾ +: PA7 -: PA2	-: PB12 -: PD15 ⁽³⁾ +: PB14 +: PD14 ⁽²⁾	+: PB0 +: PE7 ⁽²⁾ -: PB2 -: PE8 ⁽³⁾	-: PB10 -: PD13 ⁽³⁾ +: PB13 +: PD12 ⁽²⁾	+: PB11 +: PD11 -: PB15 -: PD10 ⁽³⁾	+: PC1 +: PA0 ⁽²⁾ -: PC0
Comparator outputs (motor control protection)	T1BKIN T1BKIN2 T8BKIN T8BKIN2 T1BKIN2+ T8BKIN2 TIM20BKIN TIM20BKIN2 TIM1BKIN2 + TIM8BKIN2 + TIM20BKIN2						

Table 108. Comparator input/output summary (continued)

	Comparator inputs/outputs						
	COMP1	COMP2	COMP3	COMP4	COMP5	COMP6	COMP7 ⁽³⁾
Outputs on I/Os	PA0 PF4 PA6 PA11 PB8	PA2 PA7 ⁽²⁾ PA12 PB9	PC8 PA8	PB1	PC7 PA9	PA10 PC6	PC2
Outputs to internal signals	TIM1_OCrefClear TIM1_IC1 TIM2_IC4 TIM2_OCrefClear TIM3_IC1 TIM3_OCrefClear TIM20_OCref_clr ⁽⁴⁾		TIM1_OCrefClear TIM2_OCrefClear TIM3_IC2 TIM4_IC1 TIM15_IC1 TIM15_BKIN	TIM8_OCrefClear TIM3_IC3 TIM3_OCrefClear TIM4_IC2 TIM15_OCrefClear TIM15_IC2	TIM8_OCrefClear TIM2_IC1 TIM3_OCrefClear TIM4_IC3 TIM16_BKIN TIM17_IC1	TIM8_OCrefClear TIM2_IC2 TIM2_OCrefClear TIM16_OCrefClear TIM16_IC1 TIM4_IC4	TIM1_OCrefClear TIM8_OCrefClear TIM2_IC3 TIM1_IC2 TIM17_OCrefClear TIM17_BKIN

1. Only on STM32F303x6/8 and STM32F328x8 devices
2. Only on STM32F303xB/C and STM32F358xC devices.
3. Only on STM32F303xB/C/D/E, STM32F358xC and STM32F398xE devices.
4. Only on COMP2 on STM32F303x/D/E and STM32F398xE devices.

17.3.3 COMP reset and clocks

The COMP clock provided by the clock controller is synchronous with the PCLK2 (APB2 clock).

There is no clock enable control bit provided in the RCC controller. To use a clock source for the comparator, the SYSCFG clock enable control bit must be set in the RCC controller.

Important: The polarity selection logic and the output redirection to the port works independently from the PCLK2 clock. This allows the comparator to work even in Stop mode.

17.3.4 Comparator LOCK mechanism

The comparators can be used for safety purposes, such as over-current or thermal protection. For applications having specific functional safety requirements, it is necessary to insure that the comparator programming cannot be altered in case of spurious register access or program counter corruption.

For this purpose, the comparator control and status registers can be write-protected (read-only).

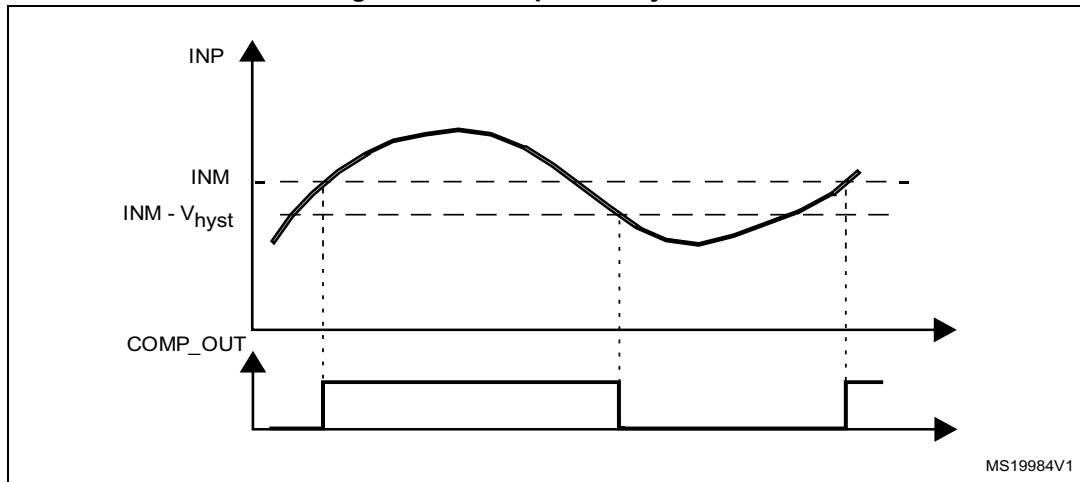
Once the programming is completed, using bits 30:0 of COMPx_CSR, the COMPx LOCK bit can be set to 1. This causes the whole COMPx_CSR register to become read-only, including the COMPx LOCK bit.

The write protection can only be reset by a MCU reset.

17.3.5 Hysteresis (on STM32F303xB/C and STM32F358xC only)

The comparator includes a programmable hysteresis to avoid spurious output transitions in case of noisy signals. The hysteresis can be disabled if it is not needed (for instance when exiting from low-power mode) to be able to force the hysteresis value using external components.

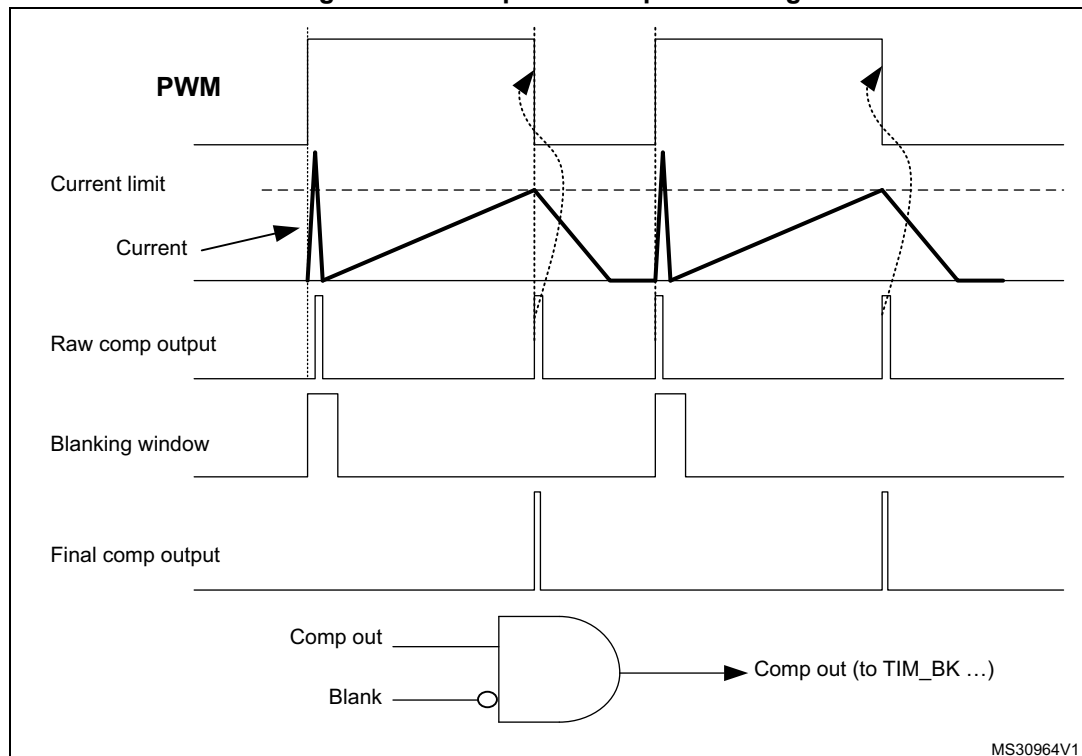
Figure 125. Comparator hysteresis



17.3.6 Comparator output blanking function

The purpose of the blanking function is to prevent the current regulation to trip upon short current spikes at the beginning of the PWM period (typically the recovery current in power switches anti parallel diodes). It consists of a selection of a blanking window which is a timer output compare signal. The selection is done by software (refer to the comparator register description for possible blanking signals). Then, the complementary of the blanking signal is ANDed with the comparator output to provide the wanted comparator output. See the example provided in the figure below.

Figure 126. Comparator output blanking



MS30964V1

17.3.7 Power mode (STM32F303xB/C and STM32F358xC only)

The comparator power consumption versus propagation delay can be adjusted to have the optimum trade-off for a given application. The bits COMPxMODE[1:0] in COMPx_CSR registers can be programmed as follows:

- 00: High speed / full power
- 01: Medium speed / medium power
- 10: Low speed / low-power
- 11: Very-low speed / ultra-low-power

17.4 COMP interrupts

The comparator outputs are internally connected to the Extended interrupts and events controller. Each comparator has its own EXTI line and can generate either interrupts or events. The same mechanism is used to exit from low-power modes.

Refer to Interrupt and events section for more details.

17.5 COMP registers

17.5.1 COMP1 control and status register (COMP1_CSR)

Note: This register is available in STM32F303xB/C/D/E, STM32F358xC and STM32F398xE only

Address offset: 0x1C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP1 LOCK	COMP1 OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP1 BLANKING			COMP1HYST [1:0] ⁽¹⁾	
rwo	r										rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP1 POL	Res.	COMP1OUTSEL				Res.	Res.	Res.	COMP1INMSEL[2:0]			COMP1MODE [1:0] ⁽¹⁾		COMP1_INP_DAC	COMP1EN
rw		rw	rw	rw	rw				rw	rw	rw	rw	rw	rw	rw

1. Only in STM32F303xB/C and STM32F358xC.

Bit 31 COMP1LOCK: Comparator 1 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

It allows to have COMP1_CSR register as read-only.

0: COMP1_CSR is read-write.

1: COMP1_CSR is read-only.

Bit 30 COMP1OUT: Comparator 1 output

This read-only bit is a copy of comparator 1 output state.

0: Output is low (non-inverting input below inverting input).

1: Output is high (non-inverting input above inverting input).

Bits 29:21 Reserved, must be kept at reset value.

Bits 20:18 **COMP1_BLANKING:** Comparator 1 blanking source

These bits select which Timer output controls the comparator 1 output blanking.

000: No blanking

001: TIM1 OC5 selected as blanking source

010: TIM2 OC3 selected as blanking source

011: TIM3 OC3 selected as blanking source

Other configurations: reserved

Bits 17:16 **COMP1HYST[1:0]** Comparator 1 hysteresis

On the STM32F303xB/C and STM32F358xC, these bits control the hysteresis level.

00: No hysteresis

01: Low hysteresis

10: Medium hysteresis

11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303xD/E and STM32F398xE, these bits are reserved and must be kept at reset value.

Bit 15 COMP1POL: Comparator 1 output polarity

This bit is used to invert the comparator 1 output.

0: Output is not inverted

1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

Bits 13:10 **COMP1OUTSEL[3:0]**: Comparator 1 output selection

These bits select which Timer input must be connected with the comparator1 output.

0000: No selection

0001: (BRK_ACTH) Timer 1 break input

0010: (BRK2) Timer 1 break input 2

0011: Timer 8 break input 1

0100: Timer 8 break input 2

0101: Timer 1 break input 2 + Timer 8 break input 2

0110: Timer 1 OCrefclear input

0111: Timer 1 input capture 1

1000: Timer 2 input capture 4

1001: Timer 2 OCrefclear input

1010: Timer 3 input capture 1

1011: Timer 3 OCrefclear input

1100: Timer 20 break input 1

1101: Timer 20 break input 2

1110: Timer 1 break input 2 + Timer 8 break input 2 + Timer 20 break input 2

1111: Reserved.

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 9:7 Reserved, must be kept at reset value.

Bits 6:4 **COMP1INMSEL[2:0]**: Comparator 1 inverting input selection

These bits allows to select the source connected to the inverting input of the comparator 1.

000: 1/4 of Vrefint

001: 1/2 of Vrefint

010: 3/4 of Vrefint

011: Vrefint

100: PA4 or DAC1 output if enabled

101: PA5 or DAC2 output if enabled

110: PA0

111: Reserved

Bits 3:2 **COMP1MODE[1:0]**: Comparator 1 mode (only in STM32F303xB/C and STM32F358xC devices)

These bits control the operating mode of the comparator1 and allows to adjust the speed/consumption.

00: High speed

01: Medium speed

10: Low-power

11: Ultra-low-power

Bit 1 **COMP1_INP_DAC**: Comparator 1 non inverting input connection to DAC output.

This bit closes a switch between comparator 1 non-inverting input (PA0) and DAC out I/O (PA4).

0: Switch open

1: Switch closed

Note: This switch is solely intended to redirect signals onto high impedance input, such as COMP1 non-inverting input (highly resistive switch).

Bit 0 **COMP1EN**: Comparator 1 enable

This bit switches COMP1 ON/OFF.

0: Comparator 1 disabled

1: Comparator 1 enabled

17.5.2 COMP2 control and status register (COMP2_CSR)

Address offset: 0x20

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP2LOCK	COMP2OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP2INMSEL[3] ⁽¹⁾	Res.	COMP2_BLANKING[2:0]			COMP2HYST[1:0] ⁽²⁾	
rw	r								rw		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP2POL	Res.	COMP2OUTSEL[3:0]				COMP2WINMODE ⁽²⁾	Res.	COMP2INPSEL ⁽²⁾	COMP2INMSEL[2:0]			COMP2MODE[1:0] ⁽²⁾		Res.	COMP2EN
rw		rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw		rw

1. Only in STM32F303x6/8 and STM32F328x8.

2. Only in STM32F303xB/C and STM32F358xC devices.

Bit 31 **COMP2LOCK**: Comparator 2 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

It allows to have COMP2_CSR register as read-only.

0: COMP2_CSR is read-write.

1: COMP2_CSR is read-only.

Bit 30 **COMP2OUT**: Comparator 2 output

This read-only bit is a copy of comparator 1 output state.

0: Output is low (non-inverting input below inverting input).

1: Output is high (non-inverting input above inverting input).

Bits 29:23 Reserved, must be kept at reset value.

Bit 22 **COMP2INMSEL[3]**: Comparator 2 inverting input selection. This bit is available only on F303x6/x8 and F328xx. It is used with Bits [6..4] to configure the Comp inverting input.

Bit 21 Reserved, must be kept at reset value.

Bits 20:18 **COMP2_BLANKING[2:0]**: Comparator 2 output blanking source

These bits select which Timer output controls the comparator 1 output blanking.

000: No blanking

001: TIM1 OC5 selected as blanking source

010: TIM2 OC3 selected as blanking source

011: TIM3 OC3 selected as blanking source

Other configurations: reserved

Bits 17:16 **COMP2HYST[1:0]**: Comparator 2 Hysteresis

On the STM32F303xB/C and STM32F358xC, these bits control the hysteresis level.

00: No hysteresis

01: Low hysteresis

10: Medium hysteresis

11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303xD/E, STM32F303x6/8, STM32F328x8 and STM32F398xx, these bits are reserved and must be kept at reset value.

- Bit 15 **COMP2POL**: Comparator 2 output polarity
 This bit is used to invert the comparator 2 output.
 0: Output is not inverted
 1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

- Bits 13:10 **COMP2OUTSEL[3:0]**: Comparator 2 output selection
 These bits select which Timer input must be connected with the comparator2 output.
 0000: No selection
 0001: (BRK_ACTH) Timer 1 break input
 0010: (BRK2) Timer 1 break input 2
 0011: (BRK_ACTH) Timer 8 break input
 0100: (BRK2) Timer 8 break input 2
 0101: Timer 1 break input2 + Timer 8 break input 2
 0110: Timer 1 OCrefclear input
 0111: Timer 1 input capture 1
 1000: Timer 2 input capture 4
 1001: Timer 2 OCrefclear input
 1010: Timer 3 input capture 1
 1011: Timer 3 OCrefclear input
 1100: Timer 20 Break Input selected
 1101: Timer 20 Break2 Input selected
 1110: Timer 1 Break2 or Timer 8 Break2 or Timer 20 Break2
 1111: Timer 20 OCrefClear Input selected

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

- Bit 9 **COMP2WINMODE**: Comparator 2 window mode (only in STM32F303xB/C and STM32F358xC devices)
 This bit selects the window mode: Both non inverting inputs of comparators share the non inverting input of Comparator 1 (PA1).
 0: Comparators 1 and 2 can not be used in window mode.
 1: Comparators 1 and 2 can be used in window mode.

Bit 8 Reserved, must be kept at reset value.

- Bit 7 **COMP2INPSEL**: Comparator 2 non inverting input selection (Only in STM32F303xB/C and STM32F358xC devices)
 0: PA7 is selected.
 1: PA3 is selected.

Note: On STM32F303x6/8, STM32F303xDxE, STM32F398xE and STM32F328, this bit is reserved. COMP2_VINP is available on PA7 whatever value is written in bit 7.

Bits 6:4 **COMP2INMSEL[2:0]**: Comparator 2 inverting input selection

These bits, together with bit 22, allows to select the source connected to the inverting input of the comparator 2.

0000: 1/4 of Vrefint

0001: 1/2 of Vrefint

0010: 3/4 of Vrefint

0011: Vrefint

0100: PA4 or DAC1_CH1 output if enabled

STM32F303xB/C/D/E, STM32F358xC and STM32F398xC:

0101: PA5 or DAC1_CH2 output if enabled

STM32F303x6/8 and STM32F328x8:

0101: DAC1_CH2 output

0110: PA2

1000 DAC2_CH1 output

Remaining combinations: reserved.

Bits 3:2 **COMP2MODE[1:0]**: Comparator 2 mode (only in STM32F303xB/C and STM32F358xC devices)

These bits control the operating mode of the comparator2 and allows to adjust the speed/consumption.

00: High speed

01: Medium speed

10: Low-power

11: Ultra-low-power

Bit 1 Reserved, must be kept at reset value.

Bit 0 **COMP2EN**: Comparator 2 enable

This bit switches COMP2 ON/OFF.

0: Comparator 2 disabled

1: Comparator 2 enabled

17.5.3 COMP3 control and status register (COMP3_CSR)

Note: This register is available in STM32F303xB/C/D/E, STM32F358xx and STM32F398xx only.

Address offset: 0x24

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP3LOCK	COMP3OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP3_BLANKING			COMP3HYST [1:0] ⁽¹⁾	
rw	r										rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP3POL	Res.	COMP3OUTSEL				Res.	Res.	COMP3INPSEL ⁽¹⁾	COMP3INMSEL[2:0]			COMP3MODE [1:0] ⁽¹⁾		Res.	COMP3EN
rw		rw	rw	rw	rw			rw	rw	rw	rw	rw	rw		rw

1. Only in STM32F303xB/C and STM32F358xC.

Bit 31 **COMP3LOCK**: Comparator 3 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.
It allows to have COMP3_CSR register as read-only.

0: COMP3_CSR is read-write.

1: COMP3_CSR is read-only.

Bit 30 **COMP3OUT**: Comparator 3 output

This read-only bit is a copy of comparator 3 output state.

0: Output is low (non-inverting input below inverting input).

1: Output is high (non-inverting input above inverting input).

Bits 29:21 Reserved, must be kept at reset value.

Bits 20:18 **COMP3_BLANKING**: Comparator 3 blanking source

These bits select which Timer output controls the comparator 3 output blanking.

000: No blanking

001: TIM1 OC5 selected as blanking source

010: Reserved.

011: TIM2 OC4 selected as blanking source

Other configurations: reserved

Bits 17:16 **COMP3HYST[1:0]** Comparator 3 hysteresis

These bits control the hysteresis level (only in STM32F303xB/C and STM32F358x).

00: No hysteresis

01: Low hysteresis

10: Medium hysteresis

11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303xD/E and STM32F398xx, these bits are reserved and must be kept at reset value.

Bit 15 **COMP3POL**: Comparator 3 output polarity

This bit is used to invert the comparator 3 output.

0: Output is not inverted

1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

Bits 13:10 **COMP3OUTSEL[3:0]**: Comparator 3 output selection

These bits are set and cleared by software if the COMP3_LOCK bit is not set.

These bits select which Timer input must be connected with the comparator 3 output.

- 0000: No timer input
- 0001: (BRK_ACTH) Timer 1 break input
- 0010: (BRK2) Timer 1 break input 2
- 0011: (BRK_ACTH) Timer 8 break input
- 0100: (BRK2) Timer 8 break input 2
- 0101: Timer 1 break input 2 or Timer 8 break input 2
- 0110: Timer 1 OCrefclear input
- 0111: Timer 4 input capture 1
- 1000: Timer 3 input capture 2
- 1001: Timer 2 OCrefclear input
- 1010: Timer 15 input capture 1
- 1011: Timer 15 break input
- 1100 = Timer 20 Break Input selected
- 1101 = Timer 20 Break2 Input selected
- 1110 = Timer 1 Break2 or Timer 8 Break2 or Timer 20 Break2

Remaining combinations: reserved.

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 9:8 Reserved, must be kept at reset value.

Bit 7 **COMP3INPSEL**: Comparator 3 non inverting input selection

- 0: PB14
- 1: PD14

Note: On STM32F303xD/E and STM32F398xE, this bit is reserved. COMP3_VINP is available on PB14.

Bits 6:4 **COMP3INMSEL[2:0]**: Comparator 1 inverting input selection

These bits allows to select the source connected to the inverting input of the comparator 3.

- 000: 1/4 of Vrefint
- 001: 1/2 of Vrefint
- 010: 3/4 of Vrefint
- 011: Vrefint
- 100: PA4 or DAC1 output if enabled
- 101: PA5 or DAC2 output if enabled
- 110: PD15
- 111: PB12

Bits 3:2 **COMP3MODE[1:0]**: Comparator 3 mode (only in STM32F303xB/C and STM32F358xC devices.)

These bits control the operating mode of the comparator 3 and allows to adjust the speed/consumption.

00: Ultra-low power

01: Low-power

10: Medium speed

11: High speed

Bit 1 Reserved, must be kept at reset value.

Bit 0 **COMP3EN**: Comparator 3 enable

This bit switches COMP3 ON/OFF.

0: Comparator 3 disabled

1: Comparator 3 enabled

17.5.4 COMP4 control and status register (COMP4_CSR)

Address offset: 0x28

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP4LOCK	COMP4OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP4INMSSEL[3] ⁽¹⁾	Res.	COMP4_BLANKING[2:0]			COMP4HYST[1:0] ⁽²⁾	
rwo	r								rw		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP4POL	Res.	COMP4OUTSEL[3:0]				COMP4WINMODE ⁽²⁾	Res.	COMP4INPSEL ⁽²⁾	COMP4INMSEL[2:0]			COMP4MODE[1:0] ⁽²⁾		Res.	COMP4EN
rw		rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw		rw

1. Only in STM32F303x6/8 and STM32F328x8.

2. Only in STM32F303xB/C and STM32F358xC.

Bit 31 **COMP4LOCK**: Comparator 4 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

It allows to have COMP4_CSR register as read-only.

0: COMP4_CSR is read-write.

1: COMP4_CSR is read-only.

Bit 30 **COMP4OUT**: Comparator 4 output

This read-only bit is a copy of comparator 4 output state.

0: Output is low (non-inverting input below inverting input).

1: Output is high (non-inverting input above inverting input).

Bits 29:23 Reserved, must be kept at reset value.

Bit 22 **COMP4INMSEL[3]**: Comparator 4 inverting input selection. This bit is available only on F303x6/x8 and F328xx. It is used with Bits [6..4] to configure the Comp inverting input.

Bit 21 Reserved, must be kept at reset value.

Bits 20:18 **COMP4_BLANKING**: Comparator 4 blanking source

These bits select which Timer output controls the comparator 4 output blanking.

- 000: No blanking
- 001: TIM3 OC4 selected as blanking source
- 010: TIM8 OC5 selected as blanking source
- 011: TIM15 OC1 selected as blanking source
- Other configurations: reserved, must be kept at reset value

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 17:16 **COMP4HYST[1:0]**: Comparator 4 Hysteresis

On the STM32F303xB/C and STM32F358xC, these bits control the hysteresis level.

- 00: No hysteresis
- 01: Low hysteresis
- 10: Medium hysteresis
- 11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303xD/E, STM32F303x6/8, STM32F398xE and STM32F328x8, these bits are reserved and must be kept at reset value.

Bit 15 **COMP4POL**: Comparator 4 output polarity

This bit is used to invert the comparator 4 output.

- 0: Output is not inverted
- 1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

Bits 13:10 **COMP4OUTSEL[3:0]**: Comparator 4 output selection

These bits select which Timer input must be connected with the comparator4 output.

- 0000: No timer input selected
- 0001: (BRK) Timer 1 break input
- 0010: (BRK2) Timer 1 break input 2
- 0011: (BRK) Timer 8 break input
- 0100: (BRK2) Timer 8 break input 2
- 0101: Timer 1 break input 2 or Timer 8 break input 2
- 0110: Timer 3 input capture 3
- 0111: Timer 8 OCrefclear input
- 1000: Timer 15 input capture 2
- 1001: Timer 4 input capture 2
- 1010: Timer 15 OCrefclear input
- 1011: Timer 3 OCrefclear input
- 1100 = Timer 20 Break Input selected
- 1101 = Timer 20 Break2 Input selected
- 1110 = Timer 1 Break2 or Timer 8 Break2 or Timer 20 Break2
- Remaining combinations: reserved.

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bit 9 **COMP4WINMODE**: Comparator 4 window mode (only in STM32F303xB/C and STM32F358xC devices)

This bit selects the window mode: both non inverting inputs comparators 3 and 4 share the non inverting input of Comparator 3 (PB14 or PD14)

0: Comparators 3 and 4 can not be used in window mode.

1: Comparators 3 and 4 can be used in window mode

Bit 8 Reserved, must be kept at reset value.

Bit 7 **COMP4INPSEL**: Comparator 4 non inverting input selection (Only on STM32F303xB/C and STM32F358xC)

0: PB0

1: PE7

Note: On STM32F303x6/8, STM32F303xDxE, STM32F398xE and STM32F328xx, this bit is reserved. COMP4_VINP is available on PB0.

Bits 6:4 **COMP4INMSEL[3:0]**: Comparator 4 inverting input selection

These bits allows to select the source connected to the inverting input of the comparator 4.

0000: 1/4 of Vrefint

0001: 1/2 of Vrefint

0010: 3/4 of Vrefint

0011: Vrefint

0100: PA4 or DAC1_CH1 output if enabled

STM32F303xB/C/D/E, STM32F358xC and STM32F398xC:

0101: PA5 or DAC1_CH2 output if enabled

STM32F303x6/8 and STM32F328xx:

0101: DAC1_CH2 output

0110: PE8

0111: PB2

1000: DAC2_CH1 output

Remaining combinations: reserved.

Bits 3:2 **COMP4MODE[1:0]**: Comparator 1 mode (only in STM32F303xB/C and STM32F358xC devices)

These bits control the operating mode of the comparator 4 and allows to adjust the speed/consumption.

00: Ultra-low-power

01: Low-power

10: Medium speed

11: High speed

Bit 1 Reserved, must be kept at reset value.

Bit 0 **COMP4EN**: Comparator 4 enable

This bit switches COMP4 ON/OFF.

0: Comparator 4 disabled

1: Comparator 4 enabled

17.5.5 COMP5 control and status register (COMP5_CSR)

Note: This register is available in STM32F303xB/C/D/E, STM32F358xx and STM32F398xx only.

Address offset: 0x2C

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP5LOCK	COMP5OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP5_BLANKING			COMP5HYST [1:0] ⁽¹⁾	
rwo	r										rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP5POL	Res.	COMP5OUTSEL				Res.	Res.	COMP5INPSEL (1)	COMP5INMSEL[2:0]			COMP5MODE [1:0] ⁽¹⁾		Res.	COMP5EN
rw		rw	rw	rw	rw			rw	rw	rw	rw	rw	rw		rw

1. Only in STM32F303xB/C.

Bit 31 COMP5LOCK: Comparator 5 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.
It allows to have COMP5_CSR register as read-only.

- 0: COMP5_CSR is read-write.
- 1: COMP5_CSR is read-only.

Bit 30 COMP5OUT: Comparator 5 output

This read-only bit is a copy of comparator 5 output state.

- 0: Output is low (non-inverting input below inverting input).
- 1: Output is high (non-inverting input above inverting input).

Bits 29:21 Reserved, must be kept at reset value.

Bits 20:18 COMP5_BLANKING: Comparator 5 blanking source

These bits select which Timer output controls the comparator 5 output blanking.

- 000: No blanking
- 001: Reserved.
- 010: TIM8 OC5 selected as blanking source
- 011: TIM3 OC3 selected as blanking source
- Other configurations: reserved

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 17:16 COMP5HYST[1:0] Comparator 5 hysteresis

These bits control the hysteresis level (only on STM32F303xB/C and STM32F358xC devices.).

- 00: No hysteresis
- 01: Low hysteresis
- 10: Medium hysteresis
- 11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303xD/E and STM32F398xE, these bits are reserved and must be kept at reset value

Bit 15 COMP5POL: Comparator 5 output polarity

This bit is used to invert the comparator 5 output.

- 0: Output is not inverted
- 1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

Bits 13:10 **COMP5OUTSEL[3:0]**: Comparator 5 output selection

These bits select which Timer input must be connected with the comparator 5 output.

0000: No timer input selected
 0001: (BRK_ACTH) Timer 1 break input
 0010: (BRK2) Timer 1 break input 2
 0011: (BRK_ACTH) Timer 8 break input
 0100: (BRK2) Timer 8 break input 2
 0101: Timer 1 break input 2 or Timer 8 break input 2
 0110: Timer 2 input capture 1
 0111: Timer 8 OCrefclear input
 1000: Timer 17 input capture 1
 1001: Timer 4 input capture 3
 1010: Timer 16 break input
 1011: Timer 3 OCrefclear input
 1100 = Timer 20 Break Input selected
 1101 = Timer 20 Break2 Input selected
 1110 = Timer 1 Break2 or Timer 8 Break2 or Timer 20 Break2
 Remaining combinations: reserved.

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 9:8 Reserved, must be kept at reset value.

Bit 7 **COMP5INPSEL**: Comparator 5 non inverting input selection

0: PD12
 1: PB13

Note: On STM32F303xDxE and STM32F398xE, this bit is reserved. COMP5_VINP is available on PB13.

Bits 6:4 **COMP5INSEL[2:0]**: Comparator 5 inverting input selection

These bits allows to select the source connected to the inverting input of the comparator 5.

000: 1/4 of Vrefint
 001: 1/2 of Vrefint
 010: 3/4 of Vrefint
 011: Vrefint
 100: PA4 or DAC1 output if enabled
 101: PA5 or DAC2 output if enabled
 110: PD13
 111: PB10

Note: On STM32F303xDxE and STM32F398xE, this bit is reserved. COMP5_VINP is available on PB13.

Bits 3:2 **COMP5MODE[1:0]**: Comparator 5 mode (Only in STM32F303xB/C and STM32F358xC devices)

These bits control the operating mode of the comparator 5 and allows to adjust the speed/consumption.

- 00: Ultra-low power
- 01: Low-power
- 10: Medium speed
- 11: High speed

Bit 1 Reserved, must be kept at reset value.

Bit 0 **COMP5EN**: Comparator 5 enable

This bit switches COMP5 ON/OFF.

- 0: Comparator 5 disabled
- 1: Comparator 5 enabled

17.5.6 COMP6 control and status register (COMP6_CSR)

Address offset: 0x30

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP6LOCK	COMP6OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP6INMS EL[3] ⁽¹⁾	Res.	COMP6_BLANKING[2:0]			COMP6HYST[1:0] ⁽²⁾	
rw	r								rw		rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP6POL	Res.	COMP6OUTSEL[3:0]				COMP6WINMODE ⁽²⁾	Res.	COMP6INPSEL ⁽²⁾	COMP6INMSEL[2:0]			COMP6MODE[1:0] ⁽²⁾		Res.	COMP6EN
rw		rw	rw	rw	rw	rw		rw	rw	rw	rw	rw	rw		rw

1. Only in STM32F303x6/8 and STM32F328.

2. Only in STM32F303xB/C and STM32F358xC devices.

Bit 31 **COMP6LOCK**: Comparator 6 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.

It allows to have COMP6_CSR register as read-only.

- 0: COMP6_CSR is read-write.
- 1: COMP6_CSR is read-only.

Bit 30 **COMP6OUT**: Comparator 6 output

This read-only bit is a copy of comparator 6 output state.

- 0: Output is low (non-inverting input below inverting input).
- 1: Output is high (non-inverting input above inverting input).

Bits 29:23 Reserved, must be kept at reset value.

Bit 22 **COMP6INMSEL[3]**: Comparator 6 inverting input selection. This bit is available only on F303x6/x8 and F328xx. It is used with Bits [6..4] to configure the Comp inverting input.

Bit 21 Reserved, must be kept at reset value.

Bits 20:18 **COMP6_BLANKING**: Comparator 6 blanking source

These bits select which Timer output controls the comparator 6 output blanking.

- 000: No blanking
- 001: Reserved
- 010: TIM8 OC5 selected as blanking source
- 011: TIM2 OC4 selected as blanking source
- 100: TIM15 OC2 selected as blanking source
- Other configurations: reserved

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

The blanking signal is active high (masking comparator output signal). It is up to the user to program the comparator and blanking signal polarity correctly.

Bits 17:16 **COMP6HYST[1:0]**: Comparator 6 Hysteresis

On the STM32F303xB/C and STM32F358xC, these bits control the hysteresis level.

- 00: No hysteresis
- 01: Low hysteresis
- 10: Medium hysteresis
- 11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303x6/8/D/E, STM32F398xE and STM32F328x8, these bits are reserved and must be kept at reset value.

Bit 15 **COMP6POL**: Comparator 6 output polarity

This bit is used to invert the comparator 6 output.

- 0: Output is not inverted
- 1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

Bits 13:10 **COMP6OUTSEL[3:0]**: Comparator 6 output selection

These bits select which Timer input must be connected with the comparator 6 output.

- 0000: No timer input
- 0001: (BRK_ACTH) Timer 1 break input
- 0010: (BRK2) Timer 1 break input 2
- 0011: (BRK_ACTH) Timer 8 break input
- 0100: (BRK2) Timer 8 break input 2
- 0101: Timer 1 break input 2 or Timer 8 break input 2
- 0110: Timer 2 input capture 2
- 0111: Timer 8 OCrefclear input
- 1000: Timer 2 OCrefclear input
- 1001: Timer 16 OCrefclear input
- 1010: Timer 16 input capture 1
- 1011: Timer 4 input capture 4
- 1100 = Timer 20 Break Input selected
- 1101 = Timer 20 Break2 Input selected
- 1110 = Timer 1 Break2 or Timer 8 Break2 or Timer 20 Break2
- Remaining combinations: reserved.

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bit 9 **COMP6WINMODE**: Comparator 6 window mode (only in STM32F303xB/C and STM32F358xC devices)

This bit selects the window mode: both non inverting inputs of comparators 6 share the non inverting input of Comparator 5 (PD12 or PB13).

0: Comparators 5 and 6 can not be used in window mode.

1: Comparators 5 and 6 can be used in window mode

Bit 8 Reserved, must be kept at reset value.

Bit 7 **COMP6INPSEL**: Comparator 6 non inverting input selection (only on STM32F303xB/C and STM32F358xx)

0: PD11

1: PB11

*Note: On STM32F303x6/x8/D/E, STM32F328x8 and STM32F398xx, this bit is reserved.
COMP6_VINP is available on PB11 whatever value is written on bit 7.*

Bits 6:4 **COMP6INMSEL[2:0]**: Comparator 6 inverting input selection

These bits allows to select the source connected to the inverting input of the comparator 6.

0000: 1/4 of Vrefint

0001: 1/2 of Vrefint

0010: 3/4 of Vrefint

0011: Vrefint

0100: PA4 or DAC1_CH1 output if enabled

STM32F303xB/C/D/E, STM32F358xC and STM32F398xC:

0101: PA5 or DAC1_CH2 output if enabled

STM32F303x6/8 and STM32F328x8:

0101: DAC1_CH2 output

0110: PD10

0111: PB15

1000: DAC2_CH1

Remaining combinations: reserved.

Bits 3:2 **COMP6MODE[1:0]**: Comparator 6 mode (only in STM32F303xB/C and STM32F358xC devices)

These bits control the operating mode of the comparator 6 and allows to adjust the speed/consumption.

00: Ultra-low-power

01: Low-power

10: Medium speed

11: High speed

Bit 1 Reserved, must be kept at reset value.

Bit 0 **COMP6EN**: Comparator 6 enable

This bit switches COMP6 ON/OFF.

0: Comparator 6 disabled

1: Comparator 6 enabled

17.5.7 COMP7 control and status register (COMP7_CSR)

Note: This register is available in STM32F303xB/C/D/E, STM32F358xx and STM32F398xx only.

Address offset: 0x34

Reset value: 0x0000 0000

31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
COMP7LOCK	COMP7OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP7_BLANKING			COMP7HYST [1:0] ⁽¹⁾	
rwo	r										rw	rw	rw	rw	rw
15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
COMP7POL	Res.	COMP7OUTSEL				Res.	Res.	COMP7INPSEL (1)	COMP7INMSEL[2:0]			COMP7MODE [1:0] ⁽¹⁾		Res.	COMP7EN
rw		rw	rw	rw	rw			rw	rw	rw	rw	rw	rw		rw

1. Only in STM32F303xB/C and STM32F358xC devices.

Bit 31 COMP7LOCK: Comparator 7 lock

This bit is write-once. It is set by software. It can only be cleared by a system reset.
It allows to have COMP7_CSR register as read-only.

- 0: COMP7_CSR is read-write.
- 1: COMP7_CSR is read-only.

Bit 30 COMP7OUT: Comparator 7 output

This read-only bit is a copy of comparator 7 output state.

- 0: Output is low (non-inverting input below inverting input).
- 1: Output is high (non-inverting input above inverting input).

Bits 29:21 Reserved, must be kept at reset value.

Bits 20:18 COMP7_BLANKING: Comparator 7 blanking source

These bits select which Timer output controls the comparator 7 output blanking.

- 000: No blanking
- 001: TIM1 OC5 selected as blanking source
- 010: TIM8 OC5 selected as blanking source
- 011: Reserved
- 100: TIM15 OC2 selected as blanking source
- Other configurations: reserved

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 17:16 COMP7HYST[1:0] Comparator 7 hysteresis

These bits control the hysteresis level. (On the STM32F303xB/C and STM32F358xC devices only.)

- 00: No hysteresis
- 01: Low hysteresis
- 10: Medium hysteresis
- 11: High hysteresis

Please refer to the electrical characteristics for the hysteresis values.

On the STM32F303xD/E and STM32F398xx, these bits are reserved and must be kept at reset value.

Bit 15 COMP7POL: Comparator 7 output polarity

This bit is used to invert the comparator 7 output.

- 0: Output is not inverted
- 1: Output is inverted

Bit 14 Reserved, must be kept at reset value.

Bits 13:10 **COMP7OUTSEL[3:0]**: Comparator 7 output selection

These bits select which Timer input must be connected with the comparator 7 output.

- 0001: (BRK) Timer 1 break input
 - 0010: (BRK2) Timer 1 break input 2
 - 0011: (BRK) Timer 8 break input
 - 0100: (BRK2) Timer 8 break input 2
 - 0101: Timer 1 break input 2 + Timer 8 break input 2
 - 0110: Timer 1 OCrefclear input
 - 0111: Timer 8 OCrefclear input
 - 1000: Timer 2 input capture 3
 - 1001: Timer 1 input capture 2
 - 1010: Timer 17 OCrefclear input
 - 1011: Timer 17 break input
 - 1100 = Timer 20 Break Input selected
 - 1101 = Timer 20 Break2 Input selected
 - 1110 = Timer 1 Break2 or Timer 8 Break2 or Timer 20 Break2
- Remaining combinations: reserved.

Note: Depending on the product, when a timer is not available, the corresponding combination is reserved.

Bits 9:8 Reserved, must be kept at reset value.

Bit 7 **COMP7INPSEL**: Comparator 7 non inverting input selection

- 0: PA0
- 1: PC1

Note: On STM32F303xD/E and STM32F398xx, this bit is reserved. COMP7_VINP is available on PC1.

Bits 6:4 **COMP7INSEL[2:0]**: Comparator 7 inverting input selection

These bits allows to select the source connected to the inverting input of the comparator 7.

- 000: 1/4 of Vrefint
- 001: 1/2 of Vrefint
- 010: 3/4 of Vrefint
- 011: Vrefint
- 100: PA4 or DAC1 output if enabled
- 101: PA5 or DAC2 output if enabled
- 110: PC0
- 111: Reserved

Bits 3:2 **COMP7MODE[1:0]**: Comparator 7 mode (Only in STM32F303xB/C and STM32F358xC devices)

These bits control the operating mode of the comparator 7 and allows to adjust the speed/consumption.

- 00: Ultra-low power
- 01: Low-power
- 10: Medium speed
- 11: High speed

Bit 1 Reserved, must be kept at reset value.

Bit 0 **COMP7EN**: Comparator 7 enable

This bit switches COMP7 ON/OFF.

- 0: Comparator 7 disabled
- 1: Comparator 7 enabled

17.5.8 COMP register map

The following table summarizes the comparator registers.

Table 109. COMP register map and reset values

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x1C	COMP1_CSR	COMP1LOCK	COMP1OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMPx_BLANKING[2:0]	COMP1HYST[1:0]	COMP1POL	Res.	COMP1OUT SEL [3:0]				Res.	Res.	Res.	COMP1INSEL[2:0]			COMP1MODE[1:0]		COMP1_INP_DAC	COMP1EN			
	Reset value	0	0														0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x20	COMP2_CSR	COMP2LOCK	COMP2OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP2INMSEL[3]	Res.	COMP2_BLANKING	Res.	COMP2POL	Res.	COMP2OUT SEL[3:0]				COMP2WINMODE	Res.	COMP2INSEL	COMP2INMSEL[2:0]			COMP2MODE[1:0]		COMP2_INP_DAC	COMP2EN			
	Reset value	0	0								0						0	0	0		0	0	0	0	0	0	0	0	0	0	0	0	0
0x24	COMP3_CSR	COMP3LOCK	COMP3OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP3_BLANKING	COMP3HYST[1:0]	COMP3POL	Res.	COMP3OUT SEL[3:0]				Res.	Res.	COMP3INSEL	COMP3INMSEL[2:0]			COMP3MODE[1:0]		Res.	COMP3EN			
	Reset value	0	0														0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x28	COMP4_CSR	COMP4LOCK	COMP4OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP4INMSEL[3]	Res.	COMP4_BLANKING	Res.	COMP4POL	Res.	COMP4OUT SEL[3:0]				COMP4WINMODE	Res.	COMP4INSEL	COMP4INMSEL[2:0]			COMP4MODE[1:0]		COMP4EN				
	Reset value	0	0								0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x2C	COMP5_CSR	COMP5LOCK	COMP5OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP5_BLANKING	COMP5HYST[1:0]	COMP5POL	Res.	COMP5OUT SEL[3:0]				Res.	Res.	COMP5INSEL	COMP5INMSEL[2:0]			COMP5MODE[1:0]		Res.	COMP5EN			
	Reset value	0	0														0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
0x30	COMP6_CSR	COMP6LOCK	COMP6OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	COMP6INMSEL[3]	Res.	COMP6_BLANKING	Res.	COMP6POL	Res.	COMP6OUT SEL[3:0]				COMP6WINMODE	Res.	COMP6INSEL	COMP6INMSEL[2:0]			COMP6MODE[1:0]		Res.	COMP6EN			
	Reset value	0	0								0						0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Table 109. COMP register map and reset values (continued)

Offset	Register	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
0x34	COMP7_CSR	COMP7LOCK	COMP7OUT	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	Res.	0	COMP7_BLANKING		0	COMP7HYST[1:0]		COMP7POL	Res.	COMP7OUT SEL[3:0]				Res.	Res.	COMP7INSEL	COMP7INMSEL[2:0]		COMP7MODE[1:0]		Res.	COMP7EN
	Reset value	0	0											0	0		0	0	0	0	0	0	0	0	0			0	0	0	0	0	0

Refer to [Section 3.2 on page 53](#) for the register boundary addresses.